

NAAN MUDALVAN PROJECT REPORT

LegalEase: AI-Powered Legal Document Generator

Preparing the Application for Local Deployment

Project Type: Artificial Intelligence / Legal Technology / Web Application

Submitted as part of the Naan Mudhalvan Project

Prepared by: _____

Register No.: _____

Department: _____

Institution: _____

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1. Abstract

Project Overview

LegalEase is an AI-powered legal document generation application designed to help users prepare structured legal documents from simple natural-language inputs. The system focuses on reducing the complexity of document preparation while keeping the workflow organized, understandable, and suitable for local deployment.

Problem Addressed

Many users find legal-document formats difficult to understand and prepare. Traditional preparation can require repeated manual typing, searching for templates, and checking whether important information has been included. LegalEase provides a guided digital workflow that converts user-provided information into a structured draft.

Proposed Solution

The application collects the required details through a simple interface, validates the input, generates a document draft using an AI component, and allows the result to be reviewed and exported. The project emphasizes usability, modular design, privacy, and local deployment.

Expected Outcome

The completed prototype demonstrates how AI can assist document preparation without replacing professional legal advice. It provides a practical academic project combining web development, artificial intelligence, database concepts, document processing, and deployment.

2. Introduction

Background

Digital transformation is changing how information-intensive tasks are performed. Legal documentation is one area where structured forms, templates, and language processing can reduce repetitive work. LegalEase explores this opportunity through an AI-assisted document generation workflow.

Project Motivation

Students and first-time users may struggle to understand the fields required in common legal documents. A guided application can present the information in a simpler sequence and generate a consistent draft from the collected inputs.

Project Scope

The project covers user input, document-type selection, validation, AI-assisted drafting, document preview, local storage or controlled export, and basic application administration. The system is intended as an educational prototype and not as a substitute for a qualified lawyer.

Project Significance

The project demonstrates a real-world use of AI in a socially relevant domain while highlighting responsible use, data protection, human review, and clear communication of limitations.

3. Existing System

Current Practice

In a conventional workflow, a user may search for a template, copy it into a word processor, manually replace placeholders, check formatting, and revise the document multiple times. This process can be slow and prone to omissions.

Limitations

Common limitations include inconsistent formatting, repeated data entry, difficulty identifying required fields, lack of guided validation, and dependency on scattered document templates.

Need for Improvement

A centralized application can provide a consistent workflow. Instead of starting from a blank document, the user can select a document type and answer guided questions.

Comparison

The proposed approach combines a structured form, reusable templates, AI-assisted language generation, validation, preview, and export in one application.

4. Proposed System

System Description

LegalEase is proposed as a modular web application. A user selects a document category, enters relevant facts, reviews the generated draft, and exports the document.

Core Workflow

The workflow consists of document selection, guided data collection, validation, prompt construction, AI generation, post-processing, preview, human review, and final export.

User-Centered Design

The interface should use clear labels, helpful examples, required-field indicators, and simple navigation. Error messages should explain what needs to be corrected rather than merely displaying technical errors.

Safety and Responsibility

Generated content should be presented as a draft. The application should avoid claiming that generated text is guaranteed to be legally correct. Sensitive data should be minimized and protected.

5. Objectives

Primary Objective

To design and develop an AI-powered application that assists users in generating structured legal-document drafts from guided inputs.

Technical Objectives

Implement a responsive user interface, backend processing, document templates, AI integration, validation, and local deployment.

Usability Objectives

Reduce repetitive manual entry, make the document workflow easier to understand, and provide a clear preview before export.

Academic Objectives

Apply concepts from artificial intelligence, web development, database management, software engineering, and deployment in one integrated project.

6. Requirements Analysis

Functional Requirements

The application should allow document-type selection, user registration or local session handling where required, guided data entry, input validation, AI-assisted generation, preview, editing, export, and basic error handling.

Non-Functional Requirements

The system should be usable, maintainable, responsive, reasonably fast, modular, secure, and suitable for local deployment.

Hardware Requirements

A development computer with a modern processor, adequate RAM, storage, and network access for installing dependencies is sufficient for the prototype. AI processing may require access to a configured model service unless a local model is used.

Software Requirements

A typical implementation may use a modern web browser, Python or Node.js backend, frontend framework, database such as SQLite, document-generation libraries, and an AI model/API or locally hosted model.

7. System Architecture

Architecture Overview

The application can be organized into presentation, application, AI, data, and document-processing layers. The presentation layer collects input and displays results. The application layer manages workflow and validation.

AI Layer

The AI layer receives carefully constructed prompts containing only the required information and returns a draft. Prompt templates should specify document structure, tone, required fields, and output constraints.

Data Layer

A database can store document metadata and application settings. Sensitive user data should be stored only when necessary and should be protected using appropriate access controls.

Document Layer

The document layer converts validated output into a consistent format such as PDF or DOCX. It can apply headings, spacing, page numbering, and other formatting rules.

8. Technology Stack

Frontend

HTML, CSS, JavaScript, or a modern framework such as React can be used to create the user interface. Responsive design allows the application to work across common screen sizes.

Backend

Python with Flask/FastAPI or Node.js with Express can provide APIs for validation, generation, and document export.

Artificial Intelligence

An appropriate language model can be used to transform structured user input into a document draft. For privacy-sensitive deployments, a locally hosted model can be considered where hardware and model requirements permit.

Database and Storage

SQLite is suitable for a lightweight local prototype. File storage can be organized into controlled application directories with clear naming and access rules.

Deployment

The application can run on localhost for demonstration. Environment variables should be used for configuration and secrets rather than hard-coding credentials.

9. Module Description

User Interface Module

Provides document selection, form fields, navigation, validation messages, preview, and export controls.

Document Template Module

Maintains reusable structures for different document categories. Templates define headings, required information, and formatting rules.

AI Generation Module

Builds prompts from validated data and requests a structured draft. Output should be checked before it is displayed to the user.

Validation Module

Checks required fields, basic formats, missing values, and consistency between related inputs.

Export Module

Converts the approved draft into a downloadable document format and applies consistent formatting.

Administration Module

Optional administrative functions can manage templates, application configuration, and system logs without exposing unnecessary user information.

10. Database Design

Purpose

A lightweight database can maintain document metadata, template information, and application settings.

Suggested Tables

Possible tables include users, document_types, templates, documents, and audit_logs. The exact schema can be simplified for an academic prototype.

Data Integrity

Primary keys, required fields, appropriate data types, and validation rules should be used. Relationships should be designed so that records can be maintained without unnecessary duplication.

Privacy

Only information necessary for the application's purpose should be retained. If documents contain sensitive information, access should be restricted and temporary processing should be preferred where practical.

11. AI Prompt and Generation Workflow

Input Preparation

The application first converts form responses into structured fields. The system should distinguish user-provided facts from instructions so the model has a clear context.

Prompt Construction

A prompt template can specify the selected document type, required sections, formatting expectations, and instructions to avoid inventing facts. The model should be told to use only the information supplied by the user.

Output Validation

Generated text should be checked for missing sections, unexpected placeholders, unsupported claims, and formatting problems before being presented.

Human Review

The user should review the draft and make corrections before final export. For documents with legal consequences, professional review may be appropriate.

12. User Interface Design

Home Screen

The home screen introduces LegalEase and provides access to document generation, saved drafts, help information, and settings.

Document Selection

Users can choose a document category from clearly labeled options. Each category can include a short explanation of the information required.

Guided Form

The form should be divided into logical sections such as parties, dates, addresses, purpose, and additional clauses. Required fields should be clearly indicated.

Preview Screen

The preview should display the generated draft in a readable format and provide actions for edit, regenerate, save, and export.

Accessibility

Readable typography, sufficient spacing, keyboard navigation, clear error messages, and meaningful labels improve accessibility and usability.

13. Preparing the Application for Local Deployment

Environment Setup

Create a dedicated project directory and virtual environment. Install the backend, frontend, database, document-processing, and AI dependencies specified by the project.

Configuration

Keep configuration values in environment variables or a local configuration file excluded from source control. API keys and passwords should never be written directly into application code.

Database Initialization

Create the local database schema and seed only the required document templates. Verify that the application can create, read, update, and export records correctly.

Run and Test

Start the backend service, open the application in a browser, create a test document, review the generated output, and verify that export works.

Local Access

For a classroom demonstration, the application can run on localhost. If access from another device is required, network and firewall settings should be configured carefully.

14. Implementation Methodology

Development Approach

An incremental approach can be used: requirements, interface prototype, backend APIs, database integration, AI integration, document export, testing, and deployment.

Version Control

Source code should be maintained in a version-control system with meaningful commits. Environment files containing secrets should be excluded.

Error Handling

The application should handle invalid inputs, unavailable AI services, malformed output, file-generation failures, and database errors gracefully.

Maintainability

Separate modules and clear function names make it easier to update templates, replace AI providers, and improve the interface without rewriting the whole system.

15. Testing

Functional Testing

Test document selection, form validation, generation, preview, editing, saving, and export.

Validation Testing

Enter missing, invalid, unusually long, and inconsistent values to confirm that the application provides appropriate feedback.

AI Output Testing

Use representative test cases and check whether required sections appear, supplied facts are preserved, and unsupported information is not introduced.

Deployment Testing

Run the application from a clean local environment and verify that all dependencies, database initialization, configuration, and document export work as expected.

User Acceptance Testing

Ask representative users to complete a document-generation task and collect feedback about clarity, navigation, and ease of use.

16. Security, Privacy and Responsible AI

Data Protection

Legal documents may contain sensitive personal information. The application should minimize data collection, restrict access, and avoid retaining information longer than necessary.

Secrets Management

Credentials and API keys must be stored securely outside source code. Access permissions should follow the principle of least privilege.

AI Limitations

AI-generated text can contain errors or inappropriate assumptions. LegalEase should therefore identify generated content as a draft and encourage verification.

Prompt Safety

Prompts should instruct the model not to invent names, dates, facts, laws, citations, or clauses that were not supported by the supplied information.

Human Oversight

The user remains responsible for reviewing the document. Where legal accuracy is important, qualified legal professionals should review the final document.

17. Advantages and Applications

Advantages

The project provides a guided workflow, reusable templates, faster drafting, consistent formatting, easier revision, and a clear path from input to export.

Educational Value

It demonstrates practical integration of AI with software engineering rather than treating AI as a standalone feature.

Potential Applications

The architecture can support educational document templates, agreements, notices, declarations, application letters, and other structured drafts, subject to appropriate legal review.

Scalability

The system can later support additional document types, multilingual interfaces, role-based access, document history, and local AI models.

18. Limitations and Future Enhancements

Current Limitations

AI-generated drafts may contain language or factual errors, and a prototype may support only a limited set of document types. Local deployment can also be constrained by available hardware and model size.

Future AI Improvements

Future versions can use retrieval-augmented generation with verified legal sources, structured output schemas, stronger validation, and domain-specific evaluation.

Product Improvements

Possible additions include multilingual support, document version history, user accounts, digital signatures where legally appropriate, template administration, and improved accessibility.

Deployment Improvements

The application can later be containerized and deployed on a controlled server or private infrastructure with monitoring, backups, authentication, and secure configuration.

19. Results and Conclusion

Expected Results

The completed prototype should demonstrate an end-to-end process in which a user enters structured information, receives an AI-assisted draft, reviews the result, and exports a formatted document.

Project Outcome

LegalEase shows how AI, web development, database concepts, document generation, and deployment can be combined into one practical application.

Conclusion

The project provides a strong technical foundation for an AI-assisted legal-document workflow while keeping human review central. Its local-deployment design also makes it suitable for classroom demonstration and controlled testing.

Key Message

LegalEase is intended to assist with document preparation, not to replace legal professionals. The value of the system comes from combining automation with validation, transparency, privacy, and human oversight.

20. References, Glossary and Viva Points

References

Use official documentation for the selected programming language, framework, database, document-generation library, and AI provider/model. Add the exact versions and URLs used in the final implementation.

Glossary

AI: Artificial Intelligence. API: Application Programming Interface. UI: User Interface. PDF: Portable Document Format. DOCX: Microsoft Word Open XML document. LLM: Large Language Model. RAG: Retrieval-Augmented Generation.

Viva Point 1

Why AI? AI can assist with transforming structured information into readable draft language, while deterministic validation controls required fields and workflow.

Viva Point 2

Why local deployment? It provides a controlled environment for demonstration, testing, and privacy-conscious processing.

Viva Point 3

What is the main limitation? Generated text must be reviewed because AI can make mistakes and cannot guarantee legal correctness.

Viva Point 4

What can be improved? Verified legal knowledge sources, stronger validation, multilingual support, better security, and professional-review workflows can be added.